MODULE 39 · FOUNDATIONS AND EXAM SKILLS - ALL DOMAINS

Practice exam 3

The final twenty questions, weighted like the real exam, with no topic repeated from papers 1 or 2.

DOMAIN Practice exam WEIGHT - MODULE 39 of 39

Treat this as the dress rehearsal

- Sit it in one go, without pausing, as you would on the day.
- Aim for **18 or more**. The real exam mixes in formats these papers can't.
- Note any question where the right answer surprised you. That's a real gap.
- After scoring, read your three domain profiles side by side.
- Don't sit the real exam on the strength of one good paper.

> ON THE EXAM
>
> Sixty questions across three papers is a sample, not a guarantee. The modules are still where the knowledge lives.

> A platform team is removing API keys from a Foundry workload. Applications have been updated to use managed identities, and roles are assigned.
>
> Keys are still enabled on the resource, and audits keep flagging them.

What should the team do next?

A Rotate the keys regularly, and store them in Key Vault.

B Delete the applications' key configuration, leaving keys enabled.

C Restrict key use to a firewall-allowed IP range instead.

D Confirm all callers use tokens, then disable key authentication.

◆ PAUSE AND DECIDE

What should the team do next?

A Rotate and vault the keys.
Better hygiene, but a valid key still bypasses RBAC entirely.

B Remove client config only.
An enabled key remains usable by anyone who has it.

C IP-restrict key use.
Narrows where a key works, not what it grants.

D **Verify callers, then disable keys.**
Microsoft's documented sequence: assign roles, verify, then turn keys off.

ON THE EXAM

Domain 1. A key grants full access with no role assignment. Identity work isn't finished until keys are off.

> Two applications share one model deployment. A nightly summarisation job sends very long prompts, and during those hours the interactive chat app receives frequent 429 responses.
>
> Neither application has changed its request volume.

What is the most likely cause and fix?

A Requests per minute exhausted; add client-side retries with backoff.

B A regional outage; fail over to a deployment in another region.

C Tokens per minute exhausted; give each workload its own deployment.

D Content filtering overload; reduce the number of enabled categories.

◆ PAUSE AND DECIDE

What is the most likely cause and fix?

A Request-rate limit.
Volume is unchanged; it's the size of the prompts that grew.

B Regional outage.
An outage wouldn't follow the nightly job's schedule.

C Token-rate limit; separate deployments.
Long prompts consume the shared token budget, throttling the neighbour.

D Content filtering.
Filtering doesn't consume throughput quota.

ON THE EXAM

Domain 1. Quota is tokens per minute as well as requests per minute. Long prompts starve their neighbours.

> An agent retrieves supplier documents and can email purchase confirmations. A supplier's document contains hidden instructions, and content moderation on the final answer sees nothing wrong.
>
> The team wants harmful or manipulated content caught earlier.

Where should guardrails be applied?

A At tool calls and tool responses, as well as input and output.

B Only on the final response, with a stricter threshold than now.

C On the user's prompt only, since that is where attacks originate.

D In the agent's instructions, telling it to ignore document commands.

◆ PAUSE AND DECIDE

Where should guardrails be applied?

A Tool call and tool response too.

Agent intervention points catch manipulated content as it enters and acts.

B Stricter output filter.

The harmful step is the tool call, which never reaches the output filter.

C Prompt only.

Misses indirect attacks, which arrive in retrieved content.

D Instructions.

Instructions are overridable, which is exactly what the attack exploits.

ON THE EXAM

Domain 1. For agents, guardrails have four intervention points, not two.

> An autonomous agent calls internal APIs overnight. Operations want to know within minutes if its tool failure rate rises above a threshold, without anyone watching a dashboard.
>
> Traces already flow to Application Insights.

What should they configure?

A A weekly scheduled evaluation against a fixed test dataset.

B An Azure Monitor log alert on a KQL query, with an action group.

C A daily review of the agent's Monitor tab by a named engineer.

D Continuous evaluation of groundedness on sampled production traffic.

◆ PAUSE AND DECIDE

What should they configure?

A Weekly evaluation.
Detects drift on test data, days after a live failure starts.

B Log alert with an action group.
Queries live telemetry on a schedule and notifies automatically.

C Daily dashboard review.
Up to a day late, and depends on a person remembering.

D Continuous groundedness.
Measures answer quality, not tool failures.

ON THE EXAM

Domain 1. 'Notify within minutes' is an alert. Evaluation measures quality; alerts watch thresholds.

> Several teams build agents in one Foundry project. An audit asks which agent accessed a storage account last month. All the agents run under the project's identity, so the logs are identical.
>
> The organisation needs per-agent attribution in future.

What should they change?

A Give each team its own subscription, to separate the audit logs.

B Add the agent's name to every request as a custom header.

C Publish each agent, so it gets its own identity.

D Enable tracing with message content recording on every agent.

◆ PAUSE AND DECIDE

What should they change?

A Subscription per team.

Separates teams, not agents, and is heavy-handed.

B Custom header.

Self-reported, and not what storage audit logs record.

C Publish agents for unique identities.

Published agents get their own identity, so actions are attributable.

D Trace message content.

Captures prompts, not which identity touched a storage account.

ON THE EXAM

Domain 1. Unpublished agents share the project identity. Publishing is what makes actions attributable.

> A team ships changes to a customer-facing agent weekly. Last week's release caused a regression that took two days to notice, and rolling back meant redeploying from source.
>
> They want faster, safer releases.

What should they adopt?

A. Manual approval of every agent response before it reaches customers.

B. A single long-lived agent, edited in place, with changes logged.

C. Separate Foundry resources for development and production.

D. Versioned agent releases, with evaluation before each promotion.

◆ PAUSE AND DECIDE

What should they adopt?

A Approve every response.
Unworkable at customer scale, and it doesn't prevent regressions.

B Edit in place.
Exactly the pattern that made rollback a redeployment.

C Separate resources.
Useful isolation, but neither versions nor gates releases.

D Versions plus evaluation gate.
Agents are named and versioned, so promotion is gated and rollback is a pointer change.

ON THE EXAM

Domain 1. Agents are versioned assets. Evaluate the new version, promote it, and keep the old one to fall back to.

> A RAG assistant sometimes answers with confident statements that don't appear in any retrieved document. Retrieval itself returns sensible passages.
>
> The team wants a metric that specifically catches invented content.

Which evaluator should they run?

A Groundedness, checking the answer against its context.

B Relevance, which checks the answer addresses the question asked.

C Fluency, which checks the answer reads naturally and correctly.

D F1 score, which compares the answer with a reference answer.

◆ PAUSE AND DECIDE

Which evaluator should they run?

A Groundedness.

Scores the answer against the retrieved context: invention is exactly what it catches.

B Relevance.

An invented answer can still be perfectly on topic.

C Fluency.

Invented text is often the most fluent text of all.

D F1 with references.

Needs reference answers, and measures overlap rather than support.

ON THE EXAM

Domain 2. Groundedness needs context; similarity needs references; relevance needs neither.

> A knowledge base of technical manuals is chunked at 2,000 tokens with no overlap. Users report that answers are often cut off mid-procedure, missing the final steps.
>
> Retrieval returns the right documents.

What should the team change first?

A Increase the number of retrieved chunks from three to twenty.

B Reduce chunk size and add overlap.

C Switch the embedding model to one with more dimensions.

D Raise the model's maximum output tokens for longer answers.

◆ PAUSE AND DECIDE

What should the team change first?

A Retrieve twenty chunks.
Floods context with mostly irrelevant text, and costs more.

B Smaller chunks with overlap.
Overlap keeps procedures intact across chunk boundaries.

C Bigger embeddings.
Retrieval already finds the right documents.

D Longer outputs.
The missing steps aren't in the context to begin with.

ON THE EXAM

Domain 2. Start around 10 percent overlap. Answers that stop mid-procedure are a chunk boundary problem.

A team upgrades their embedding model for new documents, but reindexes only documents added since the change. Search quality drops sharply, and older documents rarely appear.

Queries are embedded with the new model.

What went wrong?

A The new model produces lower quality embeddings than the old one.

B The index needs a semantic configuration to combine the two sets.

C The vector field dimensions grew, so the index tier is now too small.

D Documents and queries must use the same embedding model throughout.

◆ PAUSE AND DECIDE

What went wrong?

A Worse model.
Then new documents would suffer too, and they don't.

B Semantic configuration.
Reranks results; it can't reconcile two vector spaces.

C Index tier.
Would cause errors or cost, not silently poor matching.

D Mismatched embeddings.
Old documents sit in a different vector space from the queries.

ON THE EXAM

Domain 2. One embedding model and dimension count for documents and queries. Changing it means reindexing everything.

> A travel agent remembers each customer's seat preference and dietary needs. During testing, a customer is offered another traveller's usual seat and meal.
>
> Memory is enabled, with one scope value used for all requests.

What should the team change?

A Disable chat summary memory, keeping only user profile memory.

B Shorten the memory time to live so entries expire within hours.

C Scope memory per signed-in user.

D Move preferences into the agent's instructions instead of memory.

◆ PAUSE AND DECIDE

What should the team change?

A Disable chat summaries.

Seat and meal preferences are profile memories, which stay shared.

B Shorter TTL.

Shrinks the window; customers still see each other's data inside it.

C Per-user scope.

Scope partitions memory, so one person's facts never reach another.

D Preferences in instructions.

Static text can't hold per-customer facts at all.

ON THE EXAM

Domain 2. Scope is the keycard for memory. One shared scope is one shared notebook.

> A logistics partner runs its own agent on another cloud. It must ask your Foundry agent for delivery status, and your security team will not issue keys or add the partner to your tenant.
>
> Both sides can speak standard protocols.

What should you enable?

A Incoming A2A on your agent, so the partner's agent can call it.

B An MCP server exposing your agent's tools to the partner directly.

C A shared conversation ID that both agents append messages to.

D A webhook that pushes delivery status to the partner every hour.

◆ PAUSE AND DECIDE

What should you enable?

A Incoming A2A.

Publishes an authenticated agent endpoint another agent can call.

B Expose your tools by MCP.

Gives the partner your tools, not your agent's reasoning, and widens access.

C Shared conversation.

Conversations aren't a cross-organisation integration mechanism.

D Hourly webhook.

Pushes bulk data on a timer instead of answering questions.

ON THE EXAM

Domain 2. Agent to agent across organisations is A2A. MCP connects an agent to tools.

> A team uses a reasoning model for a classification task that is simple but extremely high volume. Costs and latency are far above budget, and accuracy is already excellent.
>
> Quality must not drop noticeably.

What should they try first?

A Keep the model, and cache responses for repeated inputs only.

B Move to a smaller non-reasoning model, then evaluate.

C Set reasoning effort to maximum, so fewer retries are needed.

D Fine-tune the reasoning model on the classification task.

◆ PAUSE AND DECIDE

What should they try first?

A Cache repeats.

Helps only where inputs repeat, and most classification inputs are new.

B Smaller model, then evaluate.

Match the model to the task, and measure whether quality actually drops.

C Maximum reasoning effort.

More reasoning tokens: slower and dearer, which is the problem.

D Fine-tune the reasoning model.

Keeps the expensive model and adds training cost.

ON THE EXAM

Domain 2. Reasoning models earn their cost on hard problems. Simple, high-volume work rarely needs them.

> A forum scans uploaded photos with Content Safety image analysis. Its code blocks an image only when the violence severity is 5 or above.
>
> Moderators find clearly graphic images, scored as medium severity, being published.

What should the team change?

A Switch to the multimodal API, which scores images on a finer scale.

B Raise the image size limit, since large photos score inconsistently.

C Block at severity 4 and above: images score only 0, 2, 4 or 6.

D Add a custom category trained on examples of graphic violence.

◆ PAUSE AND DECIDE

What should the team change?

A Multimodal API.

Adds associated text to the analysis; it doesn't fix a threshold bug.

B Size limit.

Oversized images are rejected outright, not scored inconsistently.

C Block at 4 and above.

Medium is 4, and a rule at 5 only ever blocks 6.

D Custom category.

Hours of work to detect what the built-in category already scores.

ON THE EXAM

Domain 3. Image severities come in steps of two: 0, 2, 4 and 6. A rule at 5 only ever blocks 6.

> A photographer's tool must replace only the sky in landscape photos, leaving the foreground untouched. The team sends the photo and a mask where the sky is painted white.
>
> Results alter the whole image.

What must the mask be?

A. A PNG, the photo's size, with the sky transparent.

B. A JPEG, the same size as the photo, with the sky painted black.

C. A PNG at any size, with the foreground painted a solid colour.

D. A text description of the sky region's approximate coordinates.

◆ PAUSE AND DECIDE

What must the mask be?

A PNG, same size, transparent sky.

Transparency marks the area to edit, and dimensions must match.

B JPEG with black sky.

JPEG can't store transparency, and painted colours aren't masks.

C Any size PNG.

Dimensions must match the input image exactly.

D Coordinates in text.

The edit API takes a mask file, not a description.

ON THE EXAM

Domain 3. Transparent means change this; opaque means keep this; PNG, same dimensions.

A clinic sends appointment reminders in French. Patients must be addressed formally, and job titles must agree with each practitioner's gender.

Reminders are currently translated with the default engine, and read as informal.

What should the team change?

A Train Custom Translator on ten thousand formal French sentence pairs.

B Add formal phrasing to a glossary applied to every translation.

C Translate with a language model, then post-edit the output by hand.

D Name an LLM deployment per target, setting tone and gender.

◆ PAUSE AND DECIDE

What should the team change?

A Custom Translator.
Days of training, and still no per-message gender control.

B Glossary.
Fixes specific terms, not register or grammatical agreement.

C Manual post-editing.
Solves it per message, by hand, forever.

D LLM deployment with tone and gender.
The documented controls for register and gendered agreement.

ON THE EXAM

Domain 4. Tone and gender are LLM translation controls. Plain NMT can't honour them.

> A utility wants to route incoming emails by how upset the customer sounds and how urgent the problem is. They are building this now, with no existing text analytics in place.
>
> Both signals must be reportable as fixed categories.

What should they build?

A Azure Language sentiment analysis, mapping negative scores to urgency.

B A model with a schema: enum labels for sentiment, tone and urgency.

C Key phrase extraction, with a keyword list for urgent phrases.

D Conversational language understanding, trained on labelled emails.

◆ PAUSE AND DECIDE

What should they build?

A Language sentiment.
Legacy for new builds, and negative sentiment isn't urgency.

B Model schema with enums.
Fixed categories for both signals, reportable and consistent.

C Key phrases.
Legacy, and urgency is rarely stated in one keyword.

D CLU.
Legacy, retiring by March 2029, and needs labelled training data.

ON THE EXAM

Domain 4. For new work, sentiment and tone come from a model schema. Language sentiment and CLU are legacy.

> A bank already runs a Foundry prompt agent for account queries in web chat, with tools and knowledge configured. It now wants the same agent available by phone, with natural turn-taking and interruption handling.
>
> They don't want to rebuild the agent's logic.

What should they use?

A Batch transcription of calls, feeding text to the agent afterwards.

B Real-time transcription and text to speech, wired around the agent.

C Voice Live, connected to the existing agent by name and project.

D A second agent, built specifically for voice, sharing the same tools.

◆ PAUSE AND DECIDE

What should they use?

A Batch transcription.
Asynchronous, for recordings: no live conversation at all.

B Hand-built cascade.
Workable, but they handle turn-taking, interruptions and noise themselves.

C Voice Live with the agent.
One API for the speech loop, reusing the agent as it is.

D Second voice agent.
Two agents to keep in step, for one channel.

ON THE EXAM

Domain 4. Voice Live connects to an existing agent by name; it doesn't need a separate voice agent.

> A media archive indexes press releases from Blob Storage. Old releases are deleted from storage each quarter for legal reasons, but deleted items keep appearing in search results.
>
> The indexer picks up new and edited releases correctly.

What should the team have configured?

A. A deletion detection policy, in place before the first indexer run.

B. A change detection policy, since Blob Storage has none by default.

C. A shorter indexer schedule, so deletions are noticed within minutes.

D. A skillset that marks expired documents during enrichment.

◆ PAUSE AND DECIDE

What should the team have configured?

A Deletion detection, from the start.

Deletion isn't automatic, and adding the policy later doesn't recover past deletions.

B Change detection.

Blob Storage has built-in change detection, which is why edits work.

C Shorter schedule.

More runs of a pipeline that never detects deletions.

D Skillset marking.

Enrichment can't see what's no longer there.

ON THE EXAM

Domain 5. Change detection is automatic for storage; deletion detection isn't, and it isn't retrospective.

A knowledge base returns poor answers for a common question. Investigation shows the right document ranks 90th on keyword score, and the semantic ranker is enabled with a valid configuration.

The index has vector fields populated and a vectorizer configured.

What should the team change?

A Add more fields to the semantic configuration's priority list.

B Increase the number of results returned to the application to 100.

C Lower the reranker score threshold the application filters on.

D Query hybrid, so vector recall surfaces it.

◆ PAUSE AND DECIDE

What should the team change?

A More semantic fields.

Helps the ranker judge what it receives; this document never arrives.

B Return 100 results.

Only the top 50 reach the ranker, whatever you return.

C Lower threshold.

Filters a ranked list that still lacks the document.

D **Go hybrid.**

Vector recall finds it by meaning, so the ranker can promote it.

ON THE EXAM

Domain 5. The ranker only sees the top 50. Ranked 90th is a recall problem, not a ranking one.

> A support copilot is asked questions like 'Which of our two warranty policies covers water damage after 12 months, and what's the claim limit for each?'. Single queries answer one half and miss the other.
>
> The team needs citations for each part of the answer.

What should they implement?

A. A longer system message instructing the model to answer both parts.

B. Agentic retrieval over a knowledge base, planning subqueries.

C. A larger chunk size, so both policies fit inside one chunk.

D. Two separate assistants, one per warranty policy, queried in turn.

◆ PAUSE AND DECIDE

What should they implement?

A Longer instructions.

The model can't answer from context retrieval never fetched.

B Agentic retrieval.

Plans subqueries, runs them together, reranks each, and returns references.

C Bigger chunks.

Dilutes relevance, and the policies are separate documents.

D Two assistants.

Pushes orchestration onto the user, and still merges nothing.

ON THE EXAM

Domain 5. Compound and comparative questions are what agentic retrieval's query planning is for.

How did you do?

- **Answer key:** D C A B C D, A B D C A B, C A, D B C, A D B.
- **18 or more across all three papers:** book the exam.
- **One weak domain in all three:** re-watch those modules, then re-sit paper 1.
- Mistakes that were **different each time** mean breadth, not depth. Keep revising.
- Mistakes on the **same idea** each time mean one concept, not a whole domain.

> ON THE EXAM
>
> Questions 1-6 Domain 1, 7-12 Domain 2, 13-14 Domain 3, 15-17 Domain 4, 18-20 Domain 5.

What you've built

- **Five domains**, from planning and governance to agents, vision, language and retrieval.
- **Four exam skills modules**: old versus new, confusion pairs, reading code, and mechanics.
- **Sixty practice questions**, weighted and worked, across three papers.
- Every fact checked against Microsoft's documentation, with retirements flagged as they were found.
- Check the retirement dates again the week you sit the exam. This platform moves.

> ON THE EXAM
>
> Good luck. Answer everything, read the requirements twice, and trust the tiebreakers.